Project Title: Rota and Payroll

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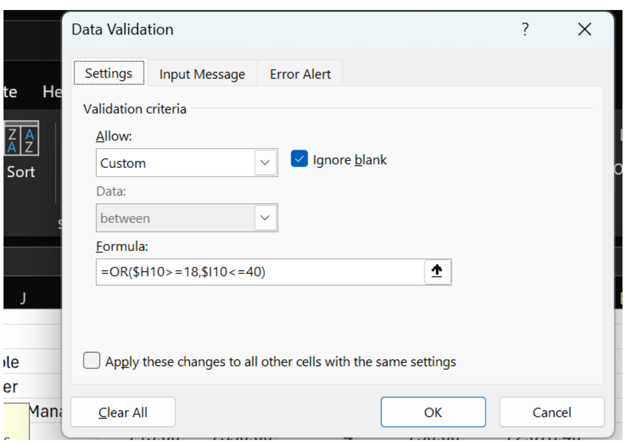
The purpose of this mini project was to design and develop a relational database system capable of storing, organizing, and analyzing Rota and payroll efficiently. The project was developed using Microsoft Excel to automate employee Rota management, payroll calculations, and data visualization.

The system was designed to improve the management of employee Rota and payroll that automatically calculates staff monthly payroll according to the hours worked by employees. The system also restricts employees younger than 18years to a maximum of 40 hours per week.

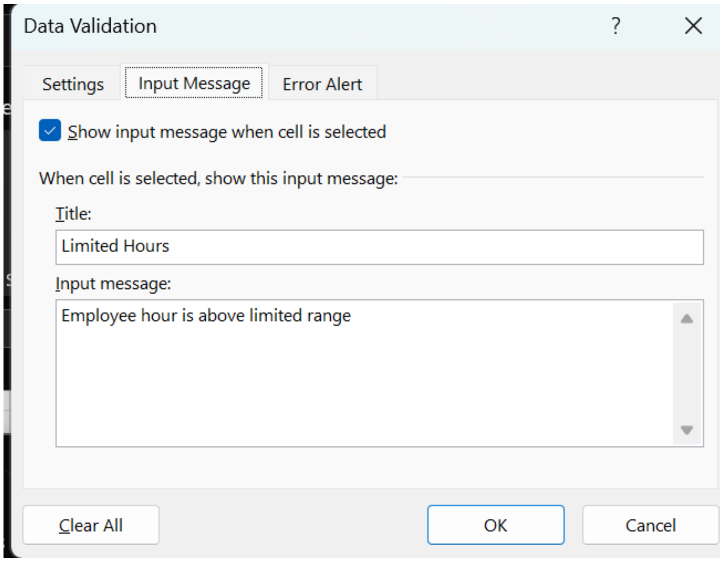
Calculation for age restriction

For staff under 18years must not be allowed to work for mor than 40 hours per week

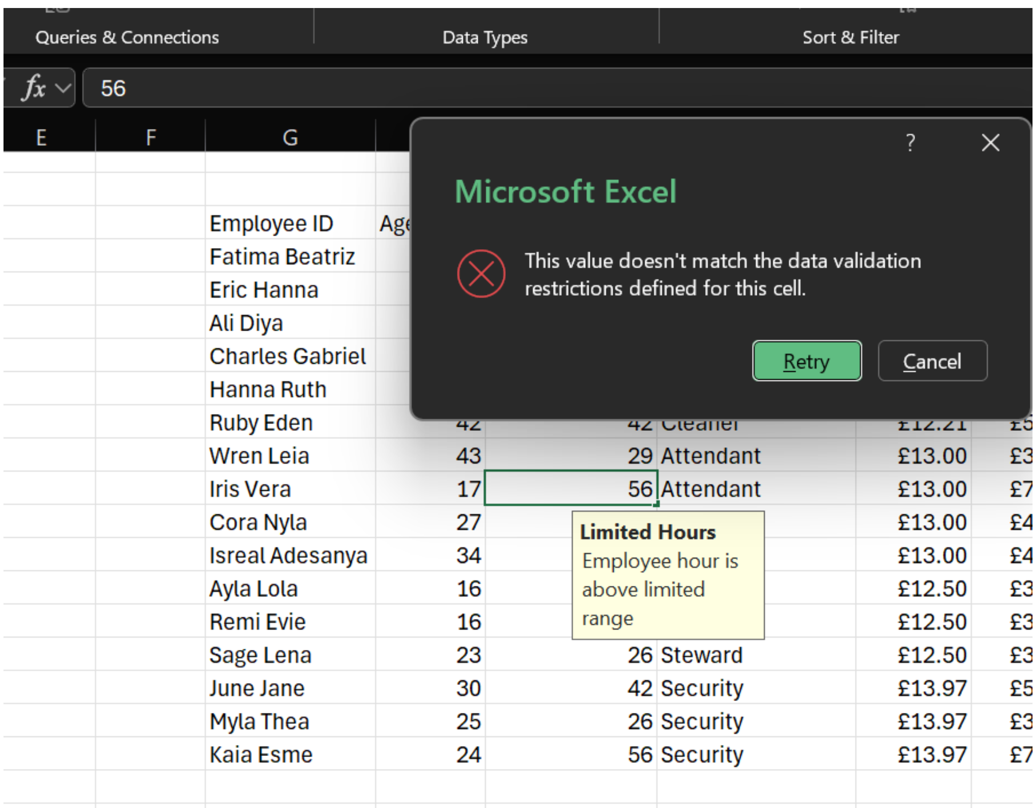
Using the data validation custom to input formular



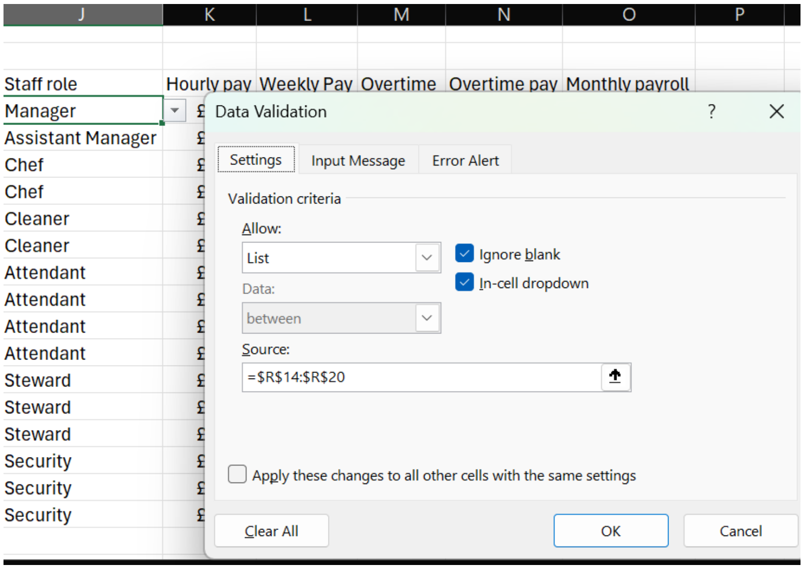
Error message display using – input message to display message when hour for employer under the age of 18 gets over 40 hours per week



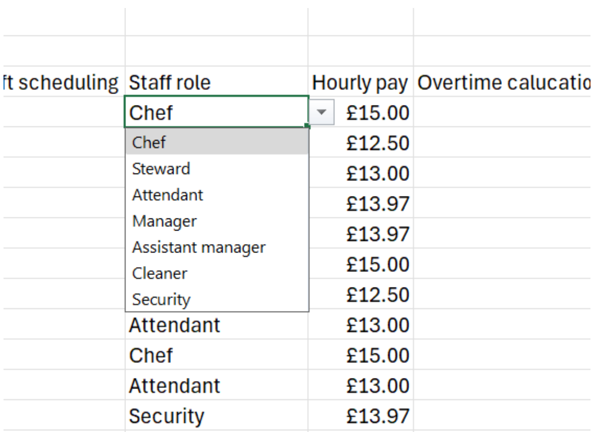
Data validation Testing



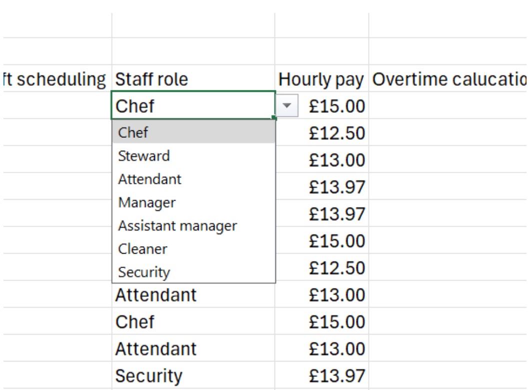
Create list data validation for staff role to choose among the hotel roles for staff



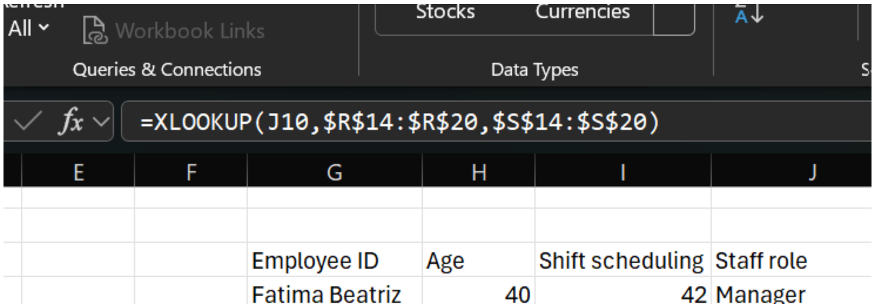
Result



Using the XLOOKUP to automatically put the hourly pay for that selected role from the roles and hourly pay data



Using XLOOKUP in the hourly pay field to automatically pick the staff role for G9 and lock in the location with $ from the data set R and S



All

Workbook Links

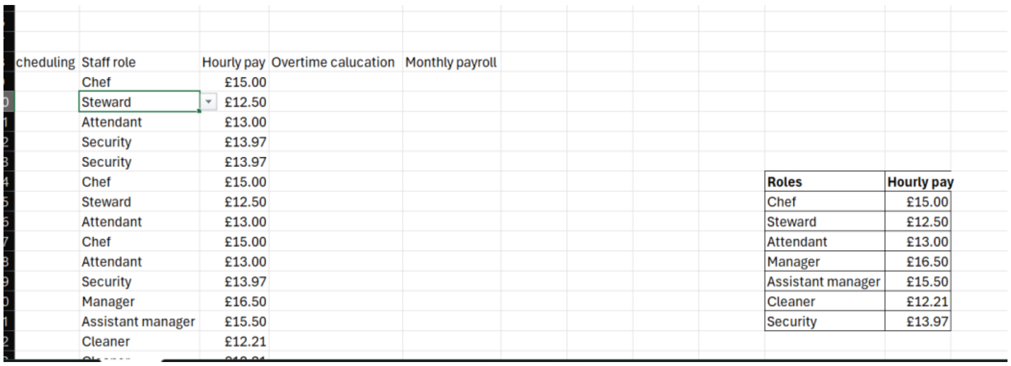
Queries & Connections

Stocks Currencies

Data Types

=XLOOKUP(J10,$R$14:$R$20,$S$14:$S$20)

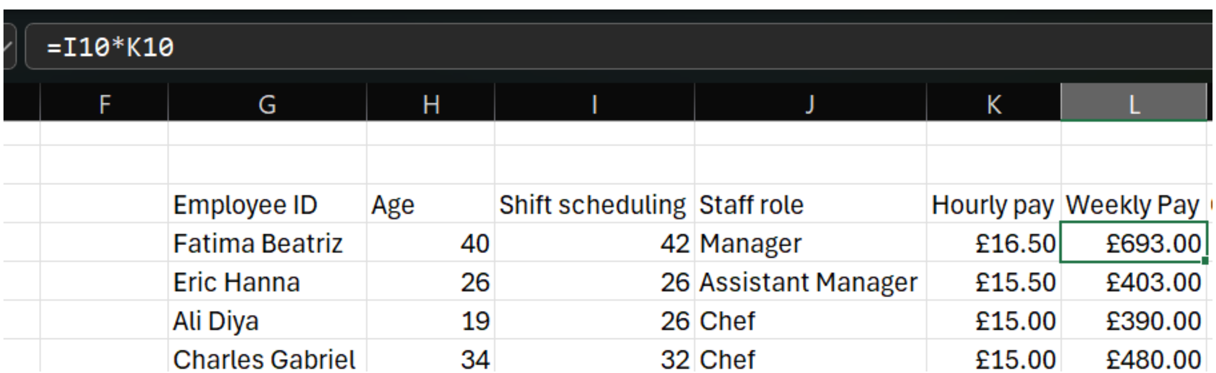
Employee ID	Age	Shift scheduling	Staff role
Fatima Beatriz	40	42	Manager



cheduling	Staff role	Hourly pay	Overtime calucation	Monthly payroll
	Chef	£15.00		
	Steward	£12.50		
	Attendant	£13.00		
	Security	£13.97		
	Security	£13.97		
	Chef	£15.00		
	Steward	£12.50		
	Attendant	£13.00		
	Chef	£15.00		
	Attendant	£13.00		
	Security	£13.97		
	Manager	£16.50		
	Assistant manager	£15.50		
	Cleaner	£12.21		

Roles	Hourly pay
Chef	£15.00
Steward	£12.50
Attendant	£13.00
Manager	£16.50
Assistant manager	£15.50
Cleaner	£12.21
Security	£13.97

Week pay calculation: Shift scheduling * Hourly pay = I10 * K10

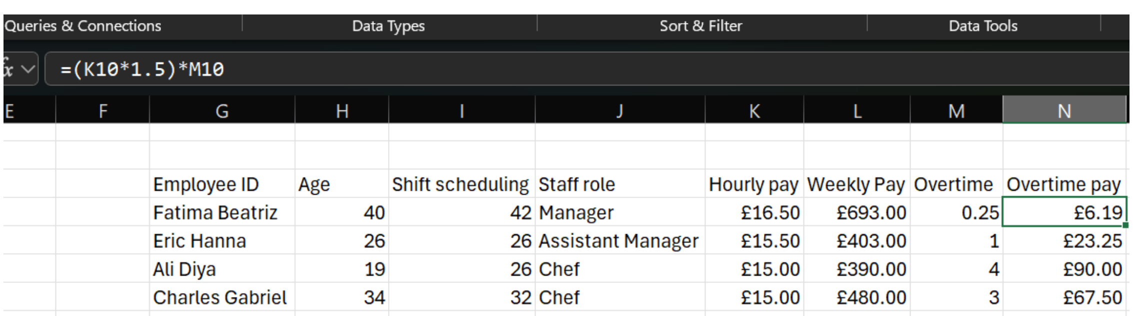


=I10*K10

Employee ID	Age	Shift scheduling	Staff role	Hourly pay	Weekly Pay
Fatima Beatriz	40	42	Manager	£16.50	£693.00
Eric Hanna	26	26	Assistant Manager	£15.50	£403.00
Ali Diya	19	26	Chef	£15.00	£390.00
Charles Gabriel	34	32	Chef	£15.00	£480.00

Overtime pay calculation: (Hourly pay * 1.5)* Overtime = (k10 * 1.5)*M10

The 1.5 is one and half per hour worked overtime for staff



Queries & Connections | Data Types | Sort & Filter | Data Tools

=(K10*1.5)*M10

Employee ID	Age	Shift scheduling	Staff role	Hourly pay	Weekly Pay	Overtime	Overtime pay
Fatima Beatriz	40	42	Manager	£16.50	£693.00	0.25	£6.19
Eric Hanna	26	26	Assistant Manager	£15.50	£403.00	1	£23.25
Ali Diya	19	26	Chef	£15.00	£390.00	4	£90.00
Charles Gabriel	34	32	Chef	£15.00	£480.00	3	£67.50
Hanna Ruth	24	26	Cleaner	£12.21	£317.46	2	£36.63

Monthly calculation: (weekly pay + Overtime pay) * 4.33 = (L10 + N10) * 4.33

If the weekly pay remains the same, so 52 weeks divided by 12 = 4.33 to calculate the monthly play

```
=(L10+N10)*4.33
```

Employee ID	Age	Shift scheduling	Staff role	Hourly pay	Weekly Pay	Overtime	Overtime pay	Monthly payroll
Fatima Beatriz	40	42	Manager	£16.50	£693.00	0.25	£6.19	£3,027.48
Eric Hanna	26	26	Assistant Manager	£15.50	£403.00	1	£23.25	£1,845.66
Ali Diya	19	26	Chef	£15.00	£390.00	4	£90.00	£2,078.40
Charles Gabriel	34	32	Chef	£15.00	£480.00	3	£67.50	£2,370.68
Hanna Ruth	24	26	Cleaner	£12.21	£317.46	2	£36.63	£1,533.21
Ruby Eden	42	26	Cleaner	£12.21	£317.46		£0.00	£1,374.60
Wren Leia	43	29	Attendant	£13.00	£377.00		£0.00	£1,632.41
Iris Vera	17	26	Attendant	£13.00	£338.00	3	£58.50	£1,716.85
Cora Nyla	27	34	Attendant	£13.00	£442.00	7	£136.50	£2,504.91
Isreal Adesanya	34	36	Attendant	£13.00	£468.00	12	£234.00	£3,039.66
Ayla Lola	16	26	Steward	£12.50	£325.00		£0.00	£1,407.25
Remi Evie	16	26	Steward	£12.50	£325.00		£0.00	£1,407.25
Sage Lena	23	26	Steward	£12.50	£325.00	2	£37.50	£1,569.63
June Jane	30	42	Security	£13.97	£586.74		£0.00	£2,540.58
Myla Thea	25	26	Security	£13.97	£363.22	2	£41.91	£1,754.21
Kaia Esme	24	56	Security	£13.97	£782.32	5	£104.78	£3,841.12

Calculation of total number of employees; using COUNTA to count because it is names and I extended the calculation to column 1054 for future input of data.

```
=COUNTA(G36:G1054)
```

Calculation for total monthly payroll; using sum function to calculate and I also extended the calculation to1015 for future input of data

```
=SUM(O36:O1015)
```

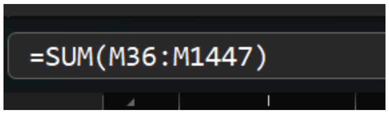
Calculation for average hourly pay; using the average function and selecting the range

```
=AVERAGE(K36:K912)
```

Calculation for highest employee paid; using DMAX function by selecting the table and field I want to check for the highest value.

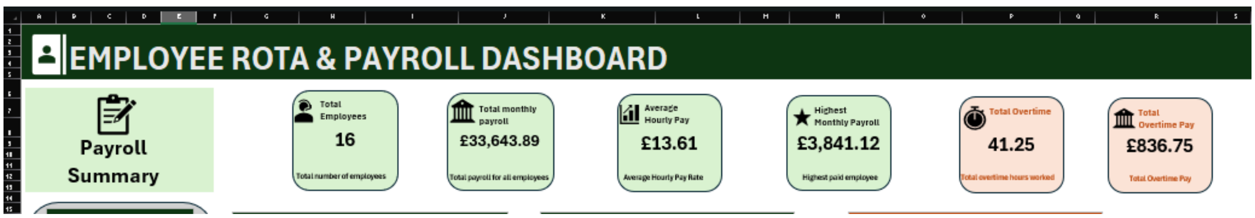
```
=DMAX(G35:O154,O35,O36:O363)
```

Calculation for total overtime worked; is calculated using the sum function



Calculation for overtime pay; using the sum function

All the KPI calculations after design

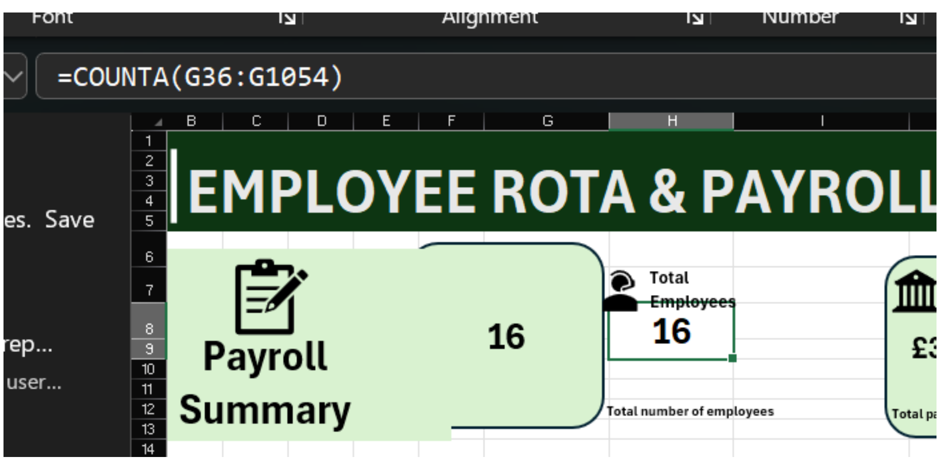


Design to enhance data visualization

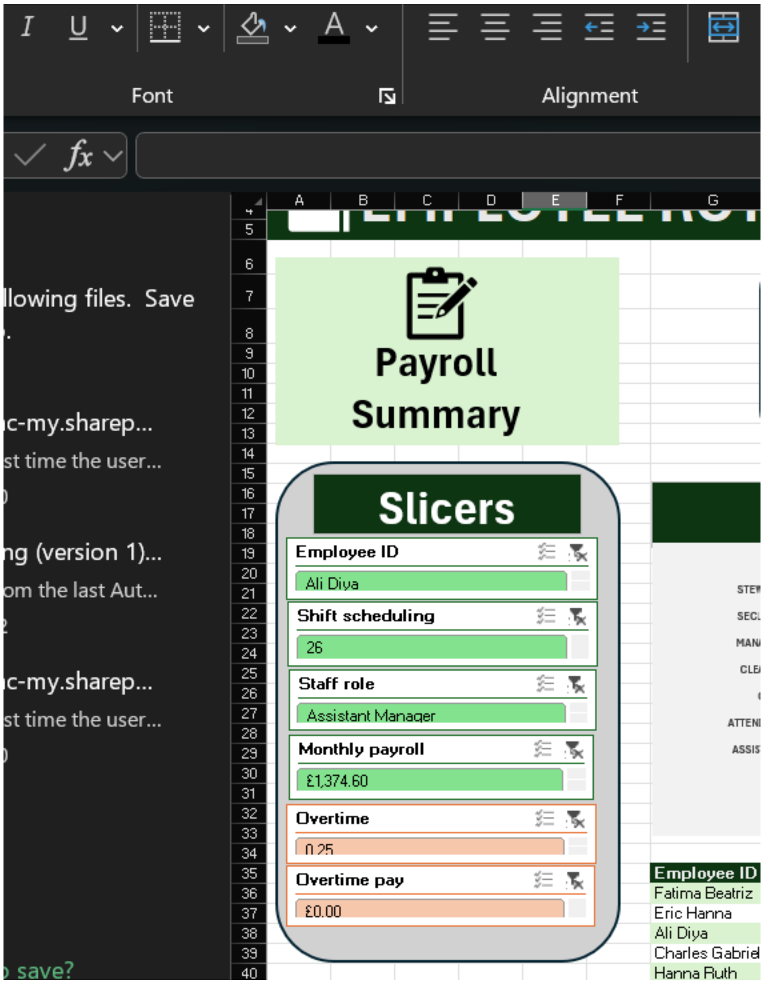
I used green for monthly payroll and orange for overtime pay. This helps improve simplicity and easy navigation in the data dashboard and charts were used for data presentation and representation.

Using shapes, icons and text to create KPI to enhance data visualization.

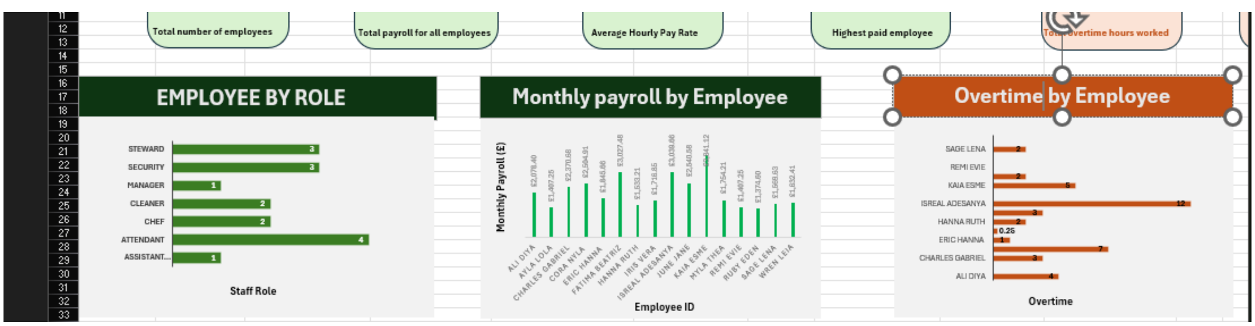
Calculations for the KPI are displayed under the shapes. The shapes cover the calculations and act like a bonnet that can be moved to make corrections or improvements of calculations.



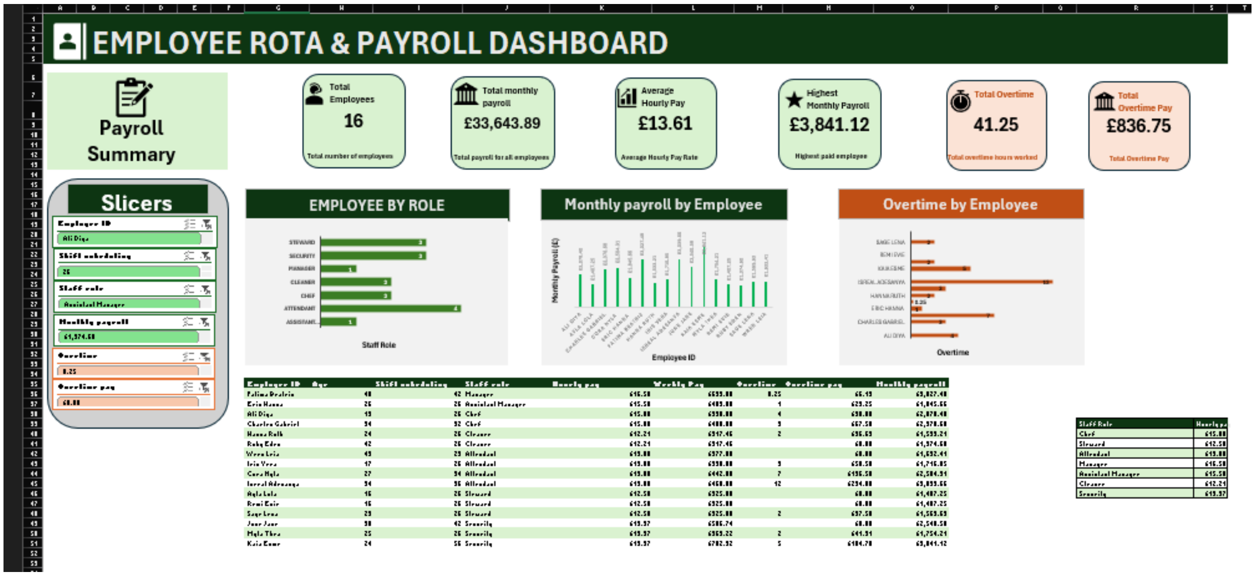
Slicer: Slicers for quick and easy access to each employee data of Rota, payroll and overtime worked at the right-hand side



Charts for employees by role, employee by monthly pay and overtime by employee



Dashboard



This project successfully demonstrates the use of Microsoft Excel as a business management tool for Rota scheduling and payroll processing. The system automates employee pay calculations, validates data entry, restricts working hours for employees under 18, and provides management with interactive dashboards and visual reports. The use of formulas, data validation, charts, slicers, and KPI indicators improves efficiency, accuracy, and decision-making. Future improvements could include automated payslip generation, leave management, and integration with employee attendance records.